

Microeconomic Foundations, I: Choice and Competitive Markets

Instructor's Manual

Chapter 5: Choice Under Uncertainty

Notes on the Chapter

Chapter 5 derives the two basic models of choice under uncertainty; that is, expected utility, first with objective probabilities (probabilities supplied in the formulation) and then with subjective probabilities (assessed by the decision maker).

- The first section of the chapter provides an introduction to the two models and the representations of preference being sought, which includes additive separability in the "state space" model.
- Section 2 concerns (objective) expected utility, relying on the Mixture-Space Theorem. It works mostly with simple (finite-support) probabilities on an arbitrary prize space, although there is some discussion of extensions to non-simple probabilities using continuity in the topology of weak convergence and the issue of bounded utility functions for "rich" spaces of probability distributions. Three points worthy of note are: (1) My mixture-space axioms are not the standard axioms (i.e., those from the original paper or from standard texts such as Fishburn, 1970); I use axioms that permit easier comparison with continuity as defined in Chapter 2 and with Savage's Sure-Thing Principle. (I do give the standard axioms as an alternative at the end of the section.) (2) In proving the Mixture-Space Theorem, I assume the existence of best and worst objects in the mixture space. Problem 5.3 (solution in the *Student's Guide*) gets the student started on the general proof. (3) Especially for students who will study finance, I believe the discussion of bounded utility is well worth tackling, at least to the point where the student understands why there is a problem and how the standard models of finance might be salvaged.
- Section 3 concerns the derivation of subjective expected utility. Savage's formulation is discussed in the introduction, along with the Sure-Thing Principle, but (in my opinion) it is too hard even for this level a book, so I rely on the derivation of Anscombe and Aumann, which mixes subjective probabilities (that is, states of nature) with objective probabilities. Two comments here: (1) I am firmly of the opinion that Anscombe and Aumann constitutes a bit of a "cheat" or (so Prof. Aumann

doesn't get mad at me) a very ingenious trick: additive separability across states of nature is, more less, a result of how they formulate things, rather than of one of their explicit axioms. I am equally firmly of the opinion that no one can say they understand A–A unless and until they understand this point, so I discuss it in the text and then have two problems, 5.4 and 5.5, to reinforce this discussion. I urge you to assign and discuss these two problems (the solutions to which are in this manual, not in the *Student's Guide*). (2) Most treatments of Anscombe and Aumann, including the original, work with a finite set of states of nature. It isn't that hard conceptually to redo their analysis with an arbitrary state space and algebra of events, although their analysis only takes you as far as finitely additive probabilities and a representation for acts that are "piecewise constant" on a finite partition of the state space. Problem 5.6 takes the student step-by-step through all this, and it shouldn't be too much of a challenge for strong students. (But it isn't as important, in my opinion, as problems 5.4 and 5.5.)

- Sections 4 and 5 round out the chapter with discussion, first of the common-prior assumption (in which I do NOT believe, so if you do, you will have to defend it—the text does not do so) and then of classic empirical and then theoretical problems with the models. The adjective "classic" in the previous sentence means that I discuss framing à la Kahnemann–Tversky, the Allais Paradox, and the Ellsberg Paradox; I do not go into more modern treatments of anomalies and alternatives to the standard models. One of the referees for the book suggested that I should "update" this discussion, including in particular reference-dependent preferences. I sympathize with the referee's desire to bring students more quickly to the forefront of current work on the subject (and his belief that this text gives an opportunity to do so), but I worry that doing this will "date" the text; I encourage instructors to bring this sort of material to the students' attention, but I've decided that, on balance, the text is better off leaving it to you.

This is a lot of material. The chapter itself is 50 pages long. I can't imagine anyone covering this chapter in one class session (or anyone who is learning this material for the first time, reading it all in one sitting). I usually take two 105-minute sessions on it and, at that, I don't have time for a serious discussion of the material in sections 4 and 5; if I had more class sessions, I'd probably stretch this chapter to three sessions.

The material at the end of Section 2 (on extending to non-simple probabilities) is only going to be accessible to students with very strong mathematical background. (It does present an opportunity for discussing how more complex spaces than R^k can be endowed with weaker and stronger topologies or, at least, what is the difference between continuity in probabilities and continuity in prizes. But I don't have that discussion here: I reserve it for Chapter 6 and a discussion of conditions under which "utility for money" will necessarily be given by a continuous function on prizes.)

Notes on the Problems

Problems 5.1 and 5.2 are straightforward.

My opinion of Problem 5.3 is not stable. The proof given in the *Guide* is a bit tricky, and students have had a hard time getting it. So I moved to a more straightforward, brute-force approach. But the straightforward and brute-force approach involves a number of cases and lots and lots of algebra. The most recent time I taught the class a student (Eric Madsen), on his own, came up with (essentially) the slicker proof. I polled his classmates, and they voted for the slick proof. So I'm back to it. I have rewritten the problem, giving a very extensive hint. And (of course) the solution is in the *Guide*. I suppose it is a good idea to assign the problem, asking students to pay attention to the hint but not to fall back on the *Guide* until they've given it a serious try on their own.

Problems 5.4 and 5.5 are, in my opinion, very helpful for understanding how Anscombe-Aumann works, and I (again) urge that they be assigned.

Problem 5.6, which shows how to do Anscombe-Aumann with an infinite state space, is far from essential and, because of the set-up required, takes almost four pages to state! So assign it with care.

Solutions to Non-Starred Problems

■ 5.2. The state-dependent subjective expected utility representation is equivalent to the additive-across-states representation. Clearly, it implies that preferences are additively separable across states. But if you have any additive representation (a representation in the form of (5.1), with state-specific utility functions u_s) and *any* (strictly positive) probability distribution p on the states, simply by defining $u'_s(x) = u_s(x)/p(s)$ gives

$$u(a) = \sum_{s \in S} u_s(a(s)) = \sum_{s \in S} p(s)u'_s(a(s));$$

the additive representation with the u_s is converted into a "state-dependent subjective expected utility representation" with p and the u'_s .

■ 5.4. The three mixture-space axioms for the mixture space Π of simple probabilities on $A = X^S$ (where S is finite) are necessary and sufficient for a representation of the form:

$$\pi \succeq \pi' \text{ if and only if } \sum_{a \in \text{supp}(\pi)} \pi(a)U(a) \geq \sum_{a \in \text{supp}(\pi')} \pi'(a)U(a),$$

where $U : A \rightarrow \mathbb{R}$ is an arbitrary function with domain A . Note that this representation is necessary and sufficient, and there is no restriction on the function U . In particular, there is no reason to suppose that U is additively separable across states of nature. So we get much less out of the three mixture-space axioms imposed on Π than we do when they are imposed on H .

Why? The text explains this at length, but the following may help your understanding of this point. Suppose that S consists of 4 states of nature, and X consists of 3 prizes. Then $A = X^S$ has $3^4 = 81$ objects, and Π is the unit simplex in R^{81} ; that is, it is a 80-dimensional set. H , on the other hand, is the four-fold product of unit simplices in R^3 ; that is, it has dimension $2^4 = 16$. That means that lots of different “members” of Π correspond to a single member of H . When we talk about preferences over Π , there is no reason that those distinct members of Π are indifferent to one another. But when we say that the decision-maker's preferences are defined over H , we are implicitly saying that she must be indifferent over them because, to her, they are the “same.” If we added to the three mixture-space axioms on Π the axiom: If π and π' are the “same” when represented as elements of H , then the decision maker is indifferent between them, we'd get back the Anscombe-Aumann representation. That fourth axiom is incredibly powerful.

■ 5.5. The setting here is: We have a finite set X of objects, and $Z = 2^X$, or the set of all subsets of X .

Now suppose we look at P , the space of all simple probabilities on Z , and we assume that the decision maker has preferences on P that satisfy the three-mixture space axioms. In terms of a representation, necessary and sufficient for the three axioms is that there exists a function $U : Z \rightarrow R$ such that the “worth” of $p \in P$ is its expected utility, computed using U . Since U can be arbitrary, this doesn't pin down $\succeq$ restricted to Z (to degenerate distributions on P) at all.

But suppose, instead, that we have preferences defined on the space $Q = [0, 1]^X$, with the interpretation mentioned in the problem statement. Some notation: For each x , let δ_x represent the element of Q that equals 1 in the x component and 0 elsewhere, and for any $q \in Q$ and $x \in X$, let q_x be the element of Q that equals $q(x)$ in component x and 0 for all other x' . Let n be the cardinality of X .

Suppose $\succeq$ on Q satisfies the three mixture-space axioms. By the Mixture-Space Theorem, there is a function $F : Q \rightarrow R$ that is linear in convex combinations and that represents $\succeq$. Normalize F so that $F((0, 0, \dots, 0)) = 0$. Define $U : X \rightarrow R$ by $U(x) = F(\delta_x)$.

Fix $x \in X$ and $a \in [0, 1]$, and (for this paragraph) let $q \in Q$ be $q(x) = a$ and $q(x') = 0$ for all $x' \neq x$. That is, q gives x with probability a and everything else with probability zero. Write $q = a\delta_x + (1 - a)(0, 0, \dots, 0)$, and apply the linearity of F to see that $F(q) = aF(\delta_x) + (1 - a)F((0, 0, \dots, 0)) = aU(x)$.

Therefore, for arbitrary $q \in Q$, $F(q_x) = q(x)U(x)$.

Now take any q , and note the identity

$$\frac{1}{n}q + \frac{n-1}{n}(0, 0, \dots, 0) = \sum_{x \in X} \frac{1}{n}q_x.$$

(Be sure you understand this identity.) Applying the linearity of F , this gives

$$\frac{1}{n} F(q) + \frac{n-1}{n} F((0, 0, \dots, 0)) = \sum_{x \in X} \frac{1}{n} F(q_x).$$

Since $F((0, 0, \dots, 0)) = 0$ and $F(q_x) = q(x)U(x)$, we can multiply through by n and the last inequality becomes

$$F(q) = \sum_{x \in X} q(x)U(x).$$

Done!

Why did this happen? Why are the mixture-space axioms applied to Q so much more powerful (in terms of the representation) than they are applied to P ? This is because in P , we see correlations between the probabilities of getting different objects, which is not at all encoded in Q . To take a concrete example, suppose $X = \{tr, tl, br, bl\}$, where tr is a right-foot tennis shoe, tl is a left-foot tennis shoe, br is a right-foot basketball shoe, and bl is a left-foot basketball shoe. Consider the following two elements of P :

$$\frac{1}{2}\{tr, tl\} + \frac{1}{2}\{br, bl\} \quad \text{and} \quad \frac{1}{2}\{br, tl\} + \frac{1}{2}\{tr, bl\}.$$

The first of these is a one-half chance of getting the two tennis shoes and a one-half chance of getting the two basketball shoes; the second is a one-half chance of getting the right-hand basketball shoe and the left-hand tennis shoe and a one-half chance of getting the left-hand basketball shoe and the right-hand tennis shoe. For most folks, the first of these would be a better lottery, because it ends up (with probability one) with a pair of matching shoes. But, as elements of Q , these are identical: each "is" a marginal probability of one-half for each type of shoe. So when we say "the decision maker has preferences on Q ," we are implicitly saying that she is indifferent between these lotteries on Z , because they are the "same" as elements of Q . That restricts her preferences quite a lot, and that is what buys for us the additively separable representation. This is precisely what happens in Anscombe-Aumann.